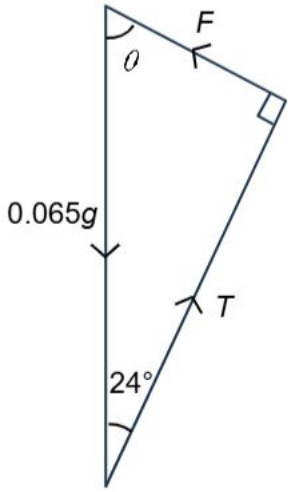
Paper 3

1 **(a)** **(i)** Horizontally: $F\sin 40° = T\sin 24°$ ……(1)

Vertically: $F\cos 40° + T\cos 24° = 0.069g$ ……(2)

Solve: $F = 0.30632 = 0.31\ \text{N}$

(ii)



F is smallest when it is perpendicular to T.

$\theta = 90° - 24° = 66°$

(b) **(i)** Since $W = F\Delta s$

$W = mg\Delta h$

$= (0.069)(9.81)(1.50 - 1.50\cos 24°)$

$= 0.087780 = 0.088\ \text{J}$

(ii) Total work done = increase in KE

$$0.0878 = \frac{p^2}{2(0.069)} - 0$$

$p = 0.11\,01$

$= 0.11\ \text{kg m s}^{-1}$

(c) **(i)** Calculate momentum p_A of sphere A after collision.

$$\frac{p_A{}^2}{2m} = 52\% \times 0.0878$$

$p_A = \sqrt{0.52 \times 0.0878 \times 2 \times 0.069)}$

$p_A = 0.079375\ \text{kg m s}^{-1}$

PCOM: $\Sigma p_i = \Sigma p_f$

$0.11 = -0.079375 + p_B$

$p_B = 0.189375 = 0.19\ \text{kg m s}^{-1}$

(ii) The statement is not valid.

If sphere A loses all its initial kinetic energy, its final momentum will be zero resulting in a smaller change in momentum for sphere A as compared to when it retained some KE and rebounded.
By the principle of conservation of momentum, the change in momentum of sphere B would be smaller. Since the final momentum of sphere B is smaller, kinetic energy of sphere B is smaller after collision.